

# Week 6 — Law of Large Numbers, Duality, Conic Programming

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Oct 3, 2026

## 1. Law of large numbers

$\tilde{\xi}_1, \tilde{\xi}_2, \tilde{\xi}_3, \dots$  : an infinite sequence of **independent and identically distributed (i.i.d.)** random variables,

each with expected value  $\mu$

**Law of large numbers** (strong law, SLLN)

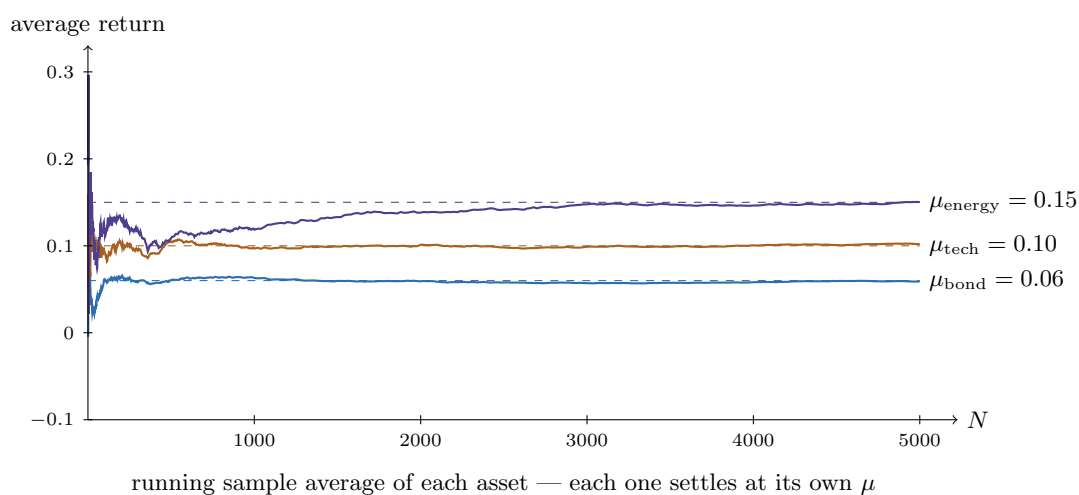
$$\mathbb{P}\left(\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i \in [N]} \tilde{\xi}_i = \mu\right) = 1$$

sample average  $\longrightarrow$  expected value

**Example** portfolio:  $\tilde{\xi}_i$  = return of an asset on day  $i$ , i.i.d. over days

| asset  | expected return $\mu$ | $\sigma$ |
|--------|-----------------------|----------|
| bond   | 0.06                  | 0.10     |
| tech   | 0.10                  | 0.20     |
| energy | 0.15                  | 0.30     |

$$\frac{1}{N} \sum_{i \in [N]} \tilde{\xi}_i = \text{average return over the first } N \text{ days}$$



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## 2. Duality theory

in this course, we focus on **convex** functions.

### Primal Problem

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & f_i(x) \leq 0 \quad \forall i \in [I] \end{aligned} \tag{P}$$

### Dual Problem

$$\begin{aligned} \max_{\theta} \min_{x \in \mathbb{R}^N} \quad & f_0(x) + \sum_{i \in [I]} \theta_i f_i(x) \\ \text{s.t.} \quad & \theta \in \mathbb{R}^I \\ & \theta \geq 0 \end{aligned} \tag{D}$$

|                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|
| $\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & f_i(x) \leq 0 \quad \forall i \in [I] \end{aligned}$ |
| $= \min_{x \in \mathbb{R}^N} f_0(x) + \underbrace{\max_{\theta \in \mathbb{R}_+^I} \sum_{i \in [I]} \theta_i f_i(x)}_{\leq 0}$           |

if  $f_i(x) \leq 0 \quad \forall i$ , then  $= 0$

if  $\exists i : f_i(x) > 0$ , then  $= +\infty$

$$\geq \max_{\theta \in \mathbb{R}_+^I} \min_{x \in \mathbb{R}^N} f_0(x) + \sum_{i \in [I]} \theta_i f_i(x)$$

### Properties:

— **weak duality**      $\min(P) \geq \max(D)$      always holds.

— **strong duality:**      $\min(P) = \max(D)$  holds

when the problem (P) is convex and the Slater's condition holds

$$\exists x \in \mathbb{R}^N, \text{ s.t. } f_i(x) < 0 \quad \forall i \in [I]$$

**In this course, we assume that strong duality holds for convex problem!!**

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### 3. Example: linear program

**Primal LP:**

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & x \in \mathbb{R}^N \\ & Ax \leq b \end{array} \quad \begin{array}{ll} f_0(x) = c^\top x \\ f_i(x) = a_i^\top x - b_i & \forall i \in [I] \end{array}$$

$$\begin{aligned} f_i(x) &\leq 0 \\ \iff a_i^\top x - b_i &\leq 0 \\ \iff a_i^\top x &\leq b_i \end{aligned}$$

**Dual LP:**

$$\begin{aligned} & \max_{\theta \in \mathbb{R}_+^I} \min_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i (a_i^\top x - b_i) \\ &= \max_{\theta \in \mathbb{R}_+^I} \min_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i a_i^\top x - \sum_{i \in [I]} \theta_i b_i \\ &= \max_{\theta \in \mathbb{R}_+^I} - \sum_{i \in [I]} \theta_i b_i + \underbrace{\min_{x \in \mathbb{R}^N} \left( c + \sum_{i \in [I]} \theta_i a_i \right)^\top x}_{= \begin{cases} 0 & \text{if } c + \sum_{i \in [I]} \theta_i a_i = 0 \\ -\infty & \text{otherwise.} \end{cases}} \\ &= \max_{\substack{\theta \in \mathbb{R}_+^I \\ A^\top \theta = -c}} -b^\top \theta \end{aligned}$$

**LP strong duality** : if there exists  $x \in \mathbb{R}^N : Ax \leq b$

$$\text{or } \theta \in \mathbb{R}^I : A^\top \theta = -c, \theta \geq 0$$

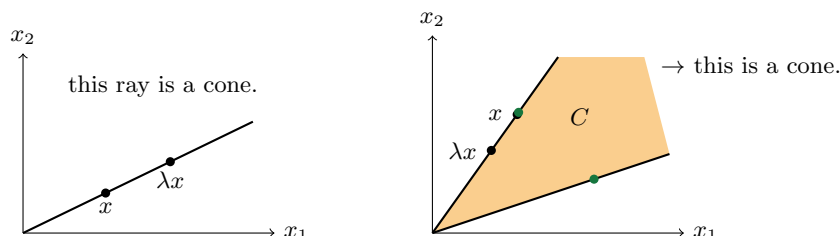
then strong duality holds

## 4. Conic programming

a conic program is a convex optimization problem.

- a **linear** objective function.
- a feasible set defines through the intersection of halfspaces  $Ax \leq b$  and a proper cone  $C$ .

**Def (cone)** a set  $C$  is a cone if  $\forall x \in C$ , we have  $\lambda x \in C \quad \forall \lambda \geq 0$



**Def (proper cone)**  $\mathbb{R}^N$  is a cone but not pointed.

$C$  is a proper cone if it's **convex, closed, pointed**

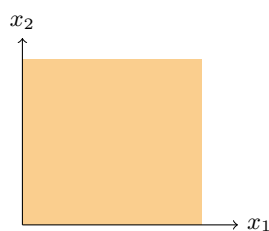
if  $x, -x \in C$  then  $x = 0$ .

**Generic conic program:**

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \leq b \\ & x \in C \end{aligned}$$

**Example:**

1) **Linear program:**  $C = \mathbb{R}_+^N$

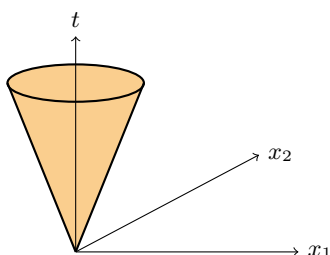


$$x \in \mathbb{R}_+^N \iff x \geq 0$$

$$\min c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0$$

2) **Second-order cone:**  $C = \text{SOC}_Q^{N+1} = \{(x, t) \in \mathbb{R}^N \times \mathbb{R} : \sqrt{x^\top Q x} \leq t\}, \quad Q \succ 0$

$Q = I$ :  $\text{SOC}^{N+1} = \{(x, t) : \|x\|_2 \leq t\}$  “ice-cream” cone



3) **Semidefinite program:**  $C = \mathbb{S}_+^K$

$\mathbb{S}^K$  : set of symmetric matrices in  $\mathbb{R}^{K \times K}$  ( $A^\top = A$ )

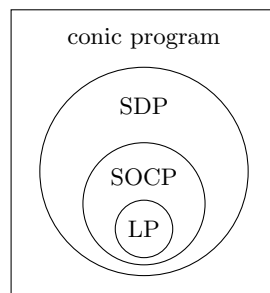
$\mathbb{S}_+^K$  : set of positive semidefinite (symmetric) matrices in  $\mathbb{R}^{K \times K}$

$$A \in \mathbb{S}_+^K \text{ if } z^\top A z \geq 0 \quad \forall z \in \mathbb{R}^K$$

**Generic semidefinite program:** the variable is a matrix  $X$

$$\begin{aligned} \min \quad & \langle D, X \rangle \\ \text{s.t.} \quad & X \in \mathbb{S}^K \\ & \langle A_i, X \rangle \leq b_i \quad \forall i \in [I] \quad \rightarrow \text{usually ignored} \\ & X \in \mathbb{S}_+^K \end{aligned}$$

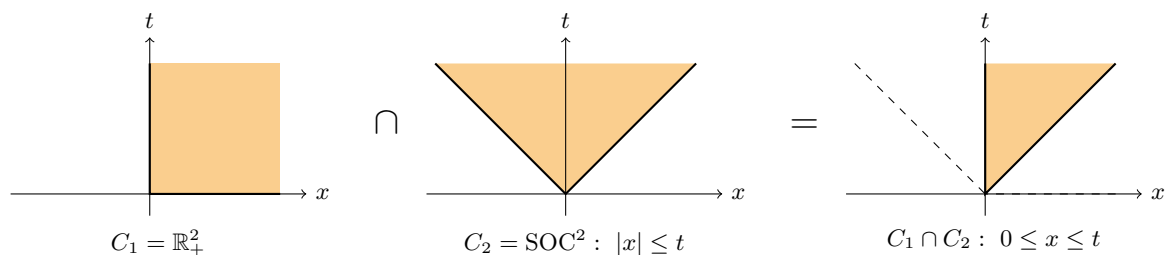
$$\langle D, X \rangle = \sum_{j,k} D_{jk} X_{jk} \text{ --- linear in } X$$



4) **Several cones at once:**

$$x \in C_1 \text{ and } x \in C_2 \iff x \in C_1 \cap C_2 \quad \text{--- again a proper cone}$$

$$x = (x^1, x^2): x^1 \in C_1 \text{ and } x^2 \in C_2 \iff x \in C_1 \times C_2 \quad \text{--- again a proper cone}$$



**Example:** portfolio, minimum variance

$$\begin{aligned} \min \quad & x^\top \Sigma x & \min \quad & t \\ \text{s.t.} \quad & x \in \mathbb{R}^N & \text{s.t.} \quad & (x, t) \in \mathbb{R}^N \times \mathbb{R} \\ & \mu^\top x \geq r & \iff & \mu^\top x \geq r, \quad e^\top x \leq 1 & \iff & A(x, t) \leq b & \text{halfspaces} \\ & e^\top x \leq 1 & & \sqrt{x^\top \Sigma x} \leq t & \iff & (x, t) \in \text{SOC}_\Sigma^{N+1} & \text{cone} \\ & x \geq 0 & & x \geq 0 & \iff & (x, t) \in \mathbb{R}_+^{N+1} & \text{cone} \end{aligned}$$

$$\Rightarrow C = \text{SOC}_\Sigma^{N+1} \cap \mathbb{R}_+^{N+1} \text{ --- a SOCP in } (x, t)$$

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## 5. Conic programming duality

**Def** (*dual cone*)

$$C^* = \{ y \in \mathbb{R}^K : y^\top x \geq 0 \quad \forall x \in C \}$$

**Self dual cones** ( $C = C^*$ )

$$— C = \mathbb{R}_+^K$$

$$— C = \text{SOC}^K \text{ (second-order cone, } Q = I) \quad \text{general } Q \succ 0: (\text{SOC}_Q^K)^* = \text{SOC}_{Q^{-1}}^K \text{ — not self-dual}$$

$$— C = \mathbb{S}_+^K$$

$$— \text{products of these: } C = C_1 \times C_2, \quad C^* = C_1^* \times C_2^* = C$$

**Primal conic program:**

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \leq b \\ & x \in C \end{aligned}$$

**Dual conic program:**

$$\begin{aligned} \max \quad & -b^\top \theta \\ \text{s.t.} \quad & \theta \in \mathbb{R}_+^I \\ & c + A^\top \theta \in C^* \end{aligned}$$

**Example 1:** linear program,  $C = \mathbb{R}_+^N = C^*$

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \leq b \\ & x \in \mathbb{R}_+^N = C \end{aligned} \quad \Longrightarrow \quad \begin{aligned} \max \quad & -b^\top \theta \\ \text{s.t.} \quad & \theta \in \mathbb{R}_+^I \\ & c + A^\top \theta \in \mathbb{R}_+^N = C^* \end{aligned}$$

$x$  free:  $C = \mathbb{R}^N$ ,  $C^* = \{0\} \Rightarrow c + A^\top \theta = 0$ , i.e.  $A^\top \theta = -c$  — the dual LP of Section 3

**Example 2:** portfolio, minimum variance — for simplicity,  $\Sigma = I$

$$\text{decision } z = \begin{bmatrix} x \\ t \end{bmatrix} \in \mathbb{R}^{N+1}, \quad c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad c^\top z = t$$

$$x \geq 0 \text{ written as halfspaces } -x \leq 0 \Rightarrow C = \text{SOC}^{N+1} = C^* \quad (\text{self-dual})$$

$$\begin{aligned} \min \quad & c^\top z \\ \text{s.t.} \quad & z \in \mathbb{R}^{N+1} \\ & \begin{bmatrix} -\mu^\top & 0 \\ e^\top & 0 \\ -I & 0 \end{bmatrix} z \leq \begin{bmatrix} -r \\ 1 \\ 0 \end{bmatrix} \quad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \lambda \end{bmatrix} \in \mathbb{R}_+^{N+2} \\ & z \in \text{SOC}^{N+1} \quad \|x\|_2 \leq t \end{aligned}$$

$$-b^\top \theta = r \theta_1 - \theta_2, \quad c + A^\top \theta = \begin{bmatrix} \theta_2 e - \theta_1 \mu - \lambda \\ 1 \end{bmatrix} \in \text{SOC}^{N+1}$$

$$\begin{aligned} \max \quad & r \theta_1 - \theta_2 \\ \text{s.t.} \quad & \theta_1, \theta_2 \geq 0, \quad \lambda \in \mathbb{R}_+^N \\ & \|\theta_2 e - \theta_1 \mu - \lambda\|_2 \leq 1 \end{aligned}$$